This report investigates the effects of home resources on academic performance, using results in mathematics, reading and science tests as a metric. Specifically, seeking to answer the question: '*Does access to computers and other educational resources at home affect pupils' learning outcomes?*'. This is done with data from the 2018 wave of PISA evaluation, which focuses on 15-year-old students in the United Kingdom. The home environment and access to educational resources outside of school are likely to contribute to academic success. The ability to use a computer and the internet, for example, offers pupils the chance to explore beyond the syllabus taught in school, further solidifying knowledge and allowing them to offer deeper analysis than their peers who are not afforded the same access. This area of investigation is of note because past studies have found that middle-class students tend to outperform their working-class peers, even when they have previously been assessed as having similar levels of intelligence, with material deprivation in the home specifically mentioned as an explanation (Douglas, 1964). This conclusion has been contested by other studies, which have instead argued that a difference in norms and values between classes serves as a better explanation for the difference in academic performance (Sugarman, 1970). This report, therefore, seeks to establish whether material differences in the home affect academic performance and, if so, to what extent. This is done by providing visualisations of the data to demonstrate distribution and the significance of individual variables' effects, as well as the use of regressions to isolate and quantify each variable's significance.

This section introduces and explains variables within the data set used in this analysis. As previously mentioned, mathematics, reading and science scores serve as dependent variables, providing a measure of academic performance. The use of all three subjects allows for more robust conclusions, as the impact of certain variables may vary across disciplines. For example, 'booksqty' is likely to have less of an impact on maths scores than it does on reading.

The independent variables used were 'desk', 'room', 'quietplace', 'homecomputer', 'homeinternet', 'books' and 'booksqty', most of which are self-explanatory, other than 'books', which establishes whether the pupil has books at home to help with their schoolwork and 'booksqty', which is the number of books in the pupil's home. This is a categorical variable done in ranges such as '0-10' and '11-25', while all other independent variables are binary, answered in either 'yes' or 'no'.

Three control variables were also implemented to mitigate the effects of omitted variable bias, these being 'gender', 'wealth' and 'hisced'. 'hisced' is an index indicating the education level of the student's highest educated parent.

Figure 1 depicts the distribution of scores across the three disciplines, all of which display approximately normal distributions- as can be seen by the bell curve overlay relative to the histogram. While there is variation within the data, there are no strongly anomalous results, with all 3 subjects peaking between 450 and 550 and demonstrating similar distribution patterns. This indicates consistent data and suggests variation is due to explanatory variables as opposed to issues with the data itself. In addition, the approximately normal distribution pattern of the data supports the later use of OLS regressions as an appropriate means to determine the significance of variables.

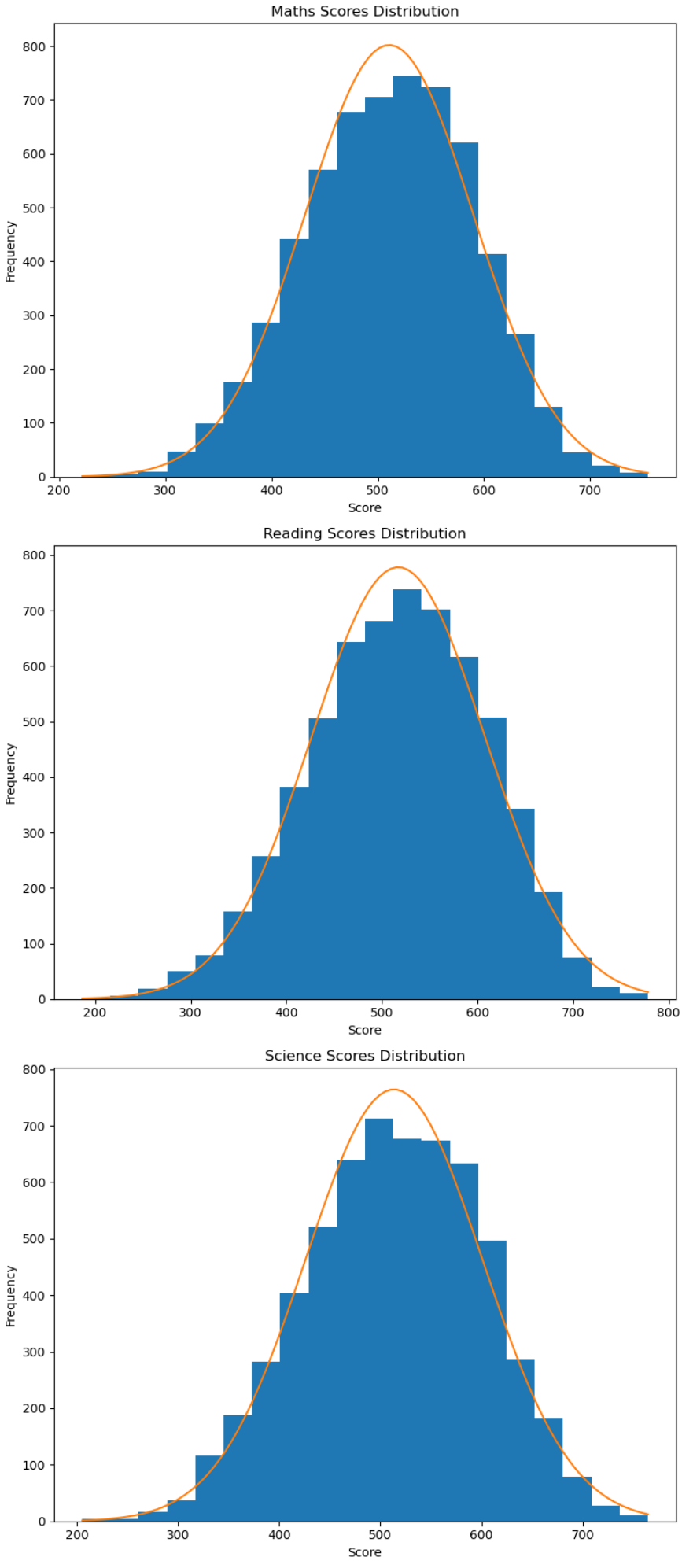


***Figure 1*: Distribution of Mathematics, Reading and Science scores**

Figure 2 is a set of boxplots demonstrating the contrast between those with and without access to a home computer across the three disciplines and between them. Pupils with access to a computer have a higher median score on tests across all three subjects, this being most noticeable in the reading results. While there is a large degree of overlap between the interquartile ranges of both the 'Yes' and 'No' groups, the distribution for students with computer access is higher in every subject.

This overlap does indicate that while computer access is a contributory factor to academic achievement, it is not a sole determinant. Overall, the boxplots suggest a positive correlation between home computer access and academic performance; however, the relationship does not appear to be particularly strong.

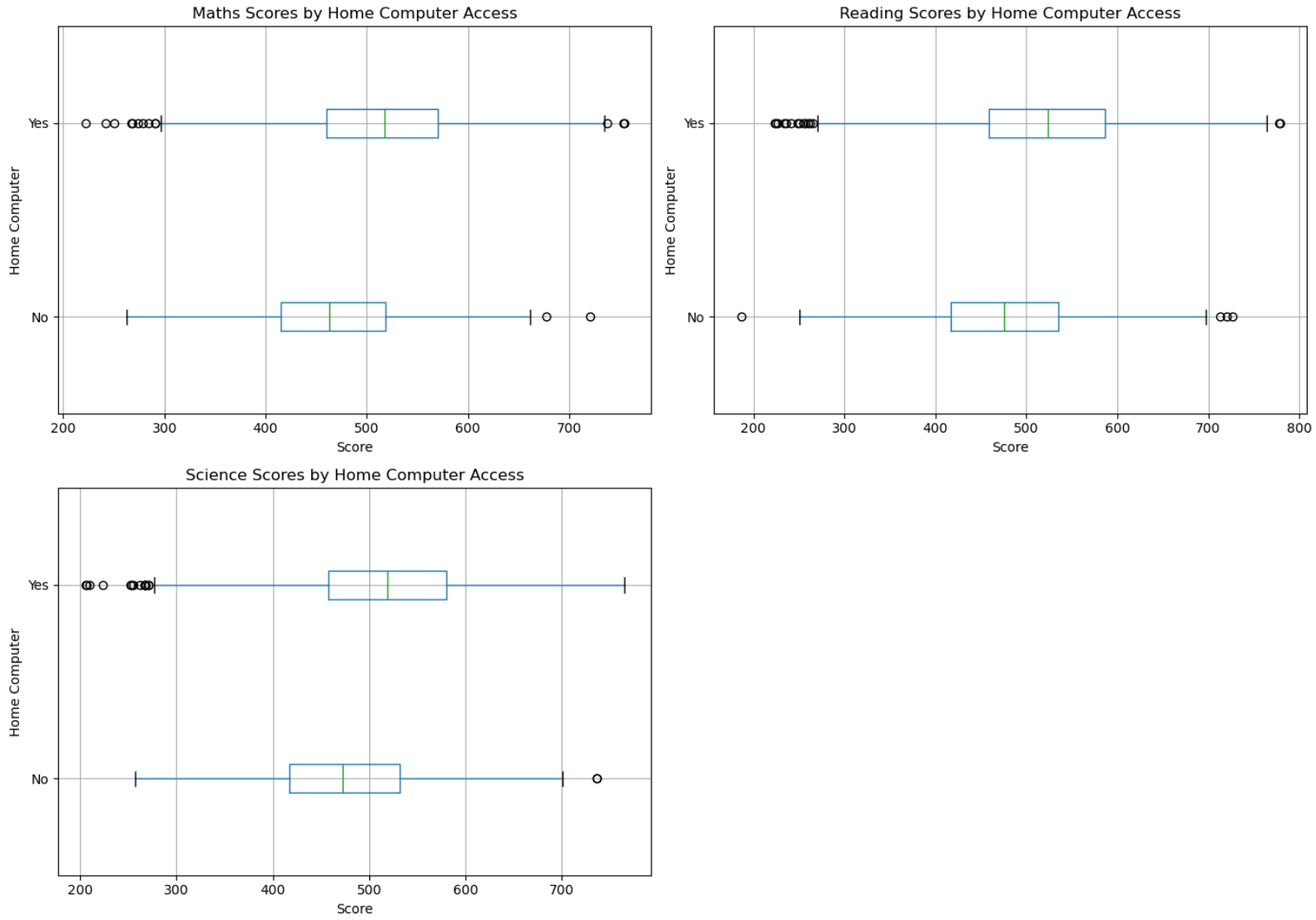


***Figure 2*: Boxplots Comparing Academic Performance Across Mathematics, Reading and Science by Access to a Home Computer**

Figure 3 depicts the relationship between access to home internet and performance across maths, reading and science. Students with internet access have a substantially higher median score (approximately 100 points higher than those without). There is a clear shift to the right in those with internet access, to the extent that there is no overlap between the interquartile ranges between the two groups in maths scores, and minimal overlap in reading and science. The effect of internet access is much more noticeable than that of computer access and is consistent across the three disciplines. This indicates that access to online learning resources is a highly influential factor in academic performance, more so than access to a home computer. It should be noted that there is a degree of limitation to this, as the total count of pupils without internet access is 46, while 5958 students do have internet access. Such a large difference in size could lead to biased conclusions, and so while there is likely a positive correlation between internet access and academic performance, this imbalance in sampling does limit reliability and results should therefore be interpreted with a degree of caution.

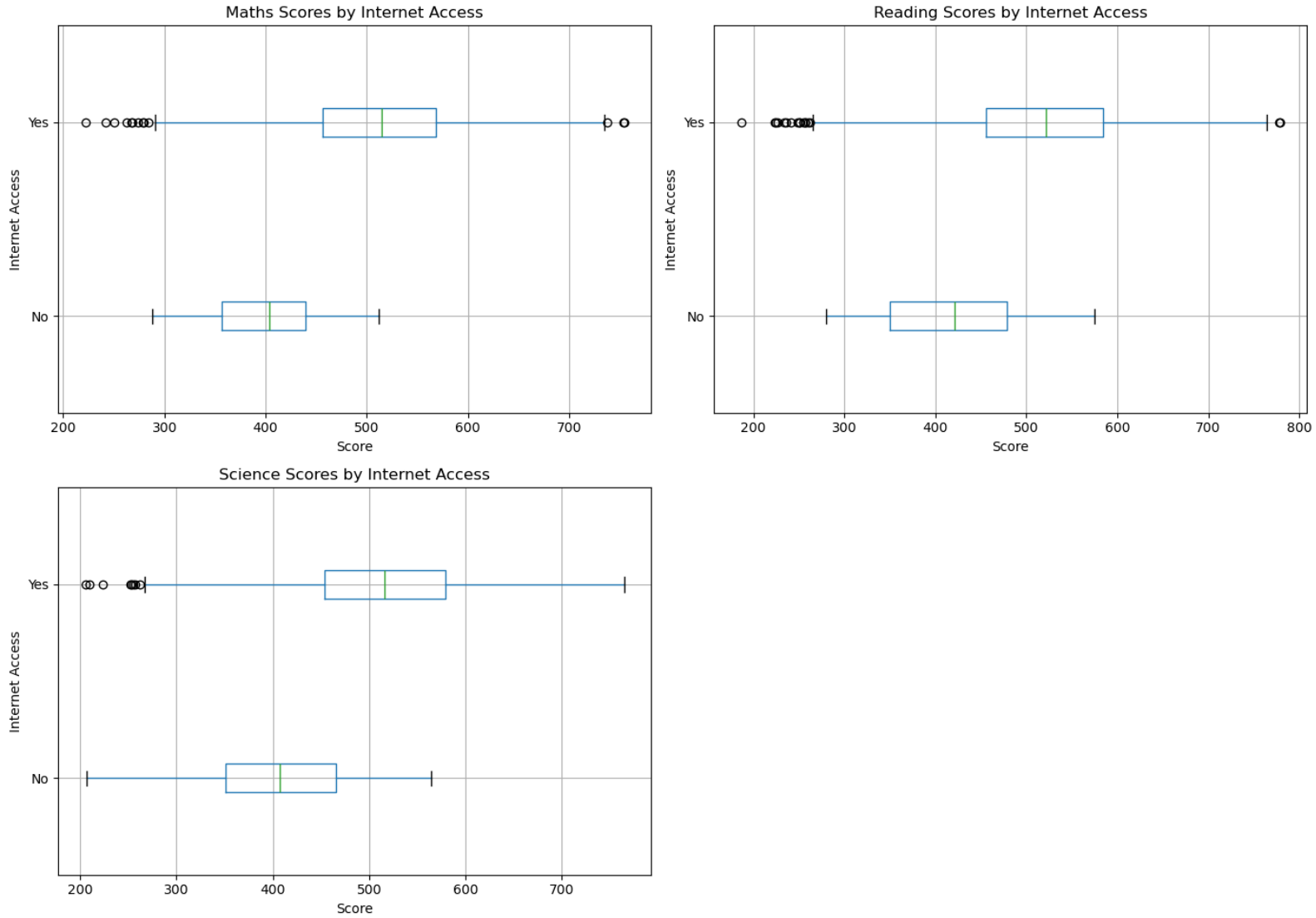


***Figure 3*: Boxplots Comparing Academic Performance Across Mathematics, Reading and Science by Access to Home Internet**

Figure 4 is a set of bar charts which demonstrates the average student performance relative to the number of books in the home. The diagrams show a clear upward trend across all three subjects, though the relationship is not perfectly linear, and it begins to plateau towards the end. There is a very clear increase in score as the number of books rises, which indicates a strong relationship between the number of books in the home and academic performance. This trend suggests that books are a significant academic resource, and proficiency in reading can aid understanding in other subjects.

The number of books in the home could, however, also be seen as a proxy variable for a family's socioeconomic status and how highly education and knowledge in general are valued within the family. While there is a clear correlation between the amount of books in the home and academic performance, this is not enough to conclude causation as, based on this data alone, it is not possible to determine whether the books themselves and the knowledge they provide are the cause for the increase in scores, or whether they reflect a broader educational environment within the home that then drives the students' achievements.

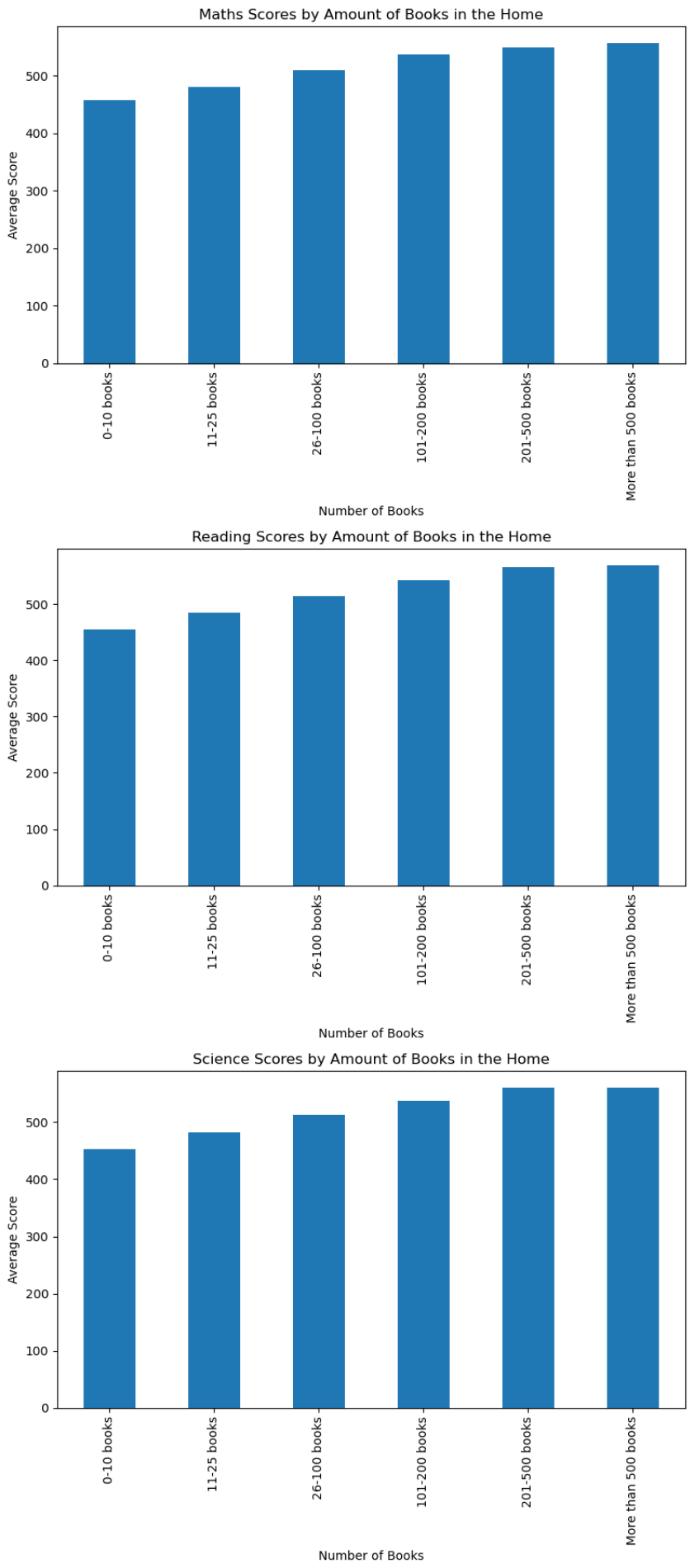


***Figure 4*: Bar Charts Demonstrating the Relationship Between Academic Performance in Mathematics, Reading and Science and the Number of Books in the Pupil's Home**

Following the visualisation and descriptive analysis of the data, three OLS regression models were used to estimate and quantify the significance of home educational resources on pupils' learning outcomes across mathematics, reading and science. Six of the independent variables were binary, and so 'Yes' was coded as one and 'No' as zero. For the categorical variable 'booksqty', dummy variables were established for five of the brackets, while the '0-10' group was omitted to serve as the baseline. Control variables were also introduced as earlier stated in the form of 'gender', 'wealth' and 'hisced' as a means to minimise omitted variable bias. Tables 1-3 present the regression results for mathematics, reading and science respectively.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	323.7965	12.023	26.932	0.000	300.228	347.365
Q("booksqty_11-25 books")[T.True]	18.6810	3.263	5.726	0.000	12.285	25.077
Q("booksqty_26-100 books")[T.True]	44.6537	2.986	14.954	0.000	38.800	50.508
Q("booksqty_101-200 books")[T.True]	70.8189	3.319	21.339	0.000	64.313	77.325
Q("booksqty_201-500 books")[T.True]	81.3146	3.532	23.024	0.000	74.391	88.238
Q("booksqty_More than 500 books")[T.True]	90.2113	4.212	21.419	0.000	81.955	98.468
desk	15.6934	3.250	4.829	0.000	9.323	22.064
room	12.2200	3.358	3.639	0.000	5.636	18.804
quietplace	-1.7196	3.402	-0.505	0.613	-8.389	4.950
homecomputer	22.0895	3.688	5.989	0.000	14.860	29.319
homeinternet	80.2430	10.894	7.366	0.000	58.887	101.599
books	14.6069	3.379	4.322	0.000	7.982	21.231
gender	-16.3720	1.837	-8.912	0.000	-19.973	-12.770
wealth	-1.7324	1.137	-1.523	0.128	-3.962	0.497
hisced	2.4817	0.823	3.017	0.003	0.869	4.094

***Table 1*: OLS Regression Results for Mathematics Scores (OECD, 2019)**

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	318.1541	13.854	22.965	0.000	290.995	345.313
Q("booksqty_11-25 books")[T.True]	26.9050	3.760	7.156	0.000	19.535	34.275
Q("booksqty_26-100 books")[T.True]	56.2203	3.441	16.339	0.000	49.475	62.966
Q("booksqty_101-200 books")[T.True]	84.2611	3.824	22.033	0.000	76.764	91.758
Q("booksqty_201-500 books")[T.True]	107.0458	4.070	26.303	0.000	99.068	115.024
Q("booksqty_More than 500 books")[T.True]	112.2331	4.853	23.126	0.000	102.719	121.747
desk	12.9731	3.744	3.465	0.001	5.633	20.314
room	16.1199	3.870	4.165	0.000	8.534	23.706
quietplace	-1.6319	3.920	-0.416	0.677	-9.317	6.053
homecomputer	20.6218	4.250	4.852	0.000	12.291	28.953
homeinternet	80.1678	12.553	6.386	0.000	55.559	104.776
books	9.2607	3.894	2.378	0.017	1.627	16.894
gender	12.9694	2.117	6.126	0.000	8.819	17.119
wealth	-9.2014	1.311	-7.021	0.000	-11.770	-6.632
hisced	1.1667	0.948	1.231	0.218	-0.691	3.025

***Table 2*: OLS Regression Results for Reading Scores (OECD, 2019)**

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	325.4549	13.384	24.317	0.000	299.218	351.692
Q("booksqty_11-25 books")[T.True]	26.8289	3.632	7.387	0.000	19.709	33.949
Q("booksqty_26-100 books")[T.True]	56.1848	3.324	16.902	0.000	49.668	62.701
Q("booksqty_101-200 books")[T.True]	80.3871	3.694	21.759	0.000	73.145	87.630
Q("booksqty_201-500 books")[T.True]	102.6104	3.932	26.099	0.000	94.903	110.318
Q("booksqty_More than 500 books")[T.True]	103.3164	4.688	22.037	0.000	94.125	112.507
desk	10.8783	3.617	3.007	0.003	3.787	17.970
room	9.9502	3.739	2.662	0.008	2.621	17.279
quietplace	-6.2152	3.787	-1.641	0.101	-13.640	1.209
homecomputer	15.5483	4.106	3.787	0.000	7.500	23.597
homeinternet	85.2220	12.127	7.027	0.000	61.449	108.995
books	10.5651	3.762	2.808	0.005	3.191	17.940
gender	-8.1233	2.045	-3.972	0.000	-12.132	-4.114
wealth	-5.1593	1.266	-4.075	0.000	-7.641	-2.677
hisced	3.1947	0.916	3.489	0.000	1.400	4.990

***Table 3*: OLS Regression Results for Science Scores (OECD, 2019)**

These regression results indicate that the home environment and educational resources within the home do correlate with academic performance across all three disciplines investigated in this report. There is variation in the size of coefficients between variables, demonstrating that they do not all contribute equally to learning outcomes. In addition, there are differences in coefficients across subjects, suggesting the impact of variables differs depending on the discipline. A clear example of this is the 'hisced' coefficient in the science regression (3.1947) compared to the reading regression (1.1667). Despite this, there are consistent results across the three models, with 'booksqty' and 'homeinternet' remaining statistically significant throughout and maintaining notably larger coefficients than other variables, indicating a stronger relationship with academic performance.

Tables 1-3 show that 'booksqty' is statistically significant across all three models, demonstrated by the consistently negligible p-values. Coefficients increase steadily across all brackets, moving from 18.68 to 90.21, 26.91 to 112.23 and 26.83 to 103.32 in maths, reading and science respectively (2 d.p). However, the increasing coefficients plateau at 'booksqty_More than 500 books', demonstrating the same relationship as depicted in earlier analysis (*Figure 4*), that being a continuous increase with a diminishing impact. These coefficients, partnered with low p-values, suggest a very strong positive correlation between the number of books in the home and learning outcomes, which then implies that access to educational materials at home aids academic performance. Books specifically could help expand a pupil's vocabulary, helping them to write in a concise and understandable manner, as well as allowing them to comprehend and communicate more advanced ideas. However, as earlier stated, books in the home could serve as a proxy variable for a home environment that prioritises and encourages academic performance, and so while there is a clear correlation, it is not enough to credit the books themselves as the cause of the academic success that they have a relationship with. Overall, there is a strong and consistent positive correlation between the amount of books in the home and academic performance, making it one of the most influential variables in the model.

Across T*ables 1-3,* the 'homeinternet' variable maintains negligible p-values, remaining statistically significant across all three models. The variable's coefficients range from 80.17 to 85.22 (2 d.p); these large and consistent values establish it as one of the most significant variables within the model. Both the lack of variation between subjects and the difference in magnitude of this variable relative to those such as 'desk' (80.17 compared to 12.97 (2 d.p) in *Table 2*) demonstrate the significance of internet connection in modern education, regardless of discipline. Such a strong correlation suggests that the role of the internet in making information and online learning resources so readily available is a huge factor in academic performance, and it therefore serves as one of the strongest predictors for a pupil's learning outcomes. These results highlight the same dynamic depicted in *Figure 3*, which suggests a higher academic performance among pupils with internet access. However, this regression data is subject to the same limitation as *Figure 3*, that being the small sample size of the 'No' group is small relative to the 'Yes' group (46 to 5958). The lack of certainty in the exact value this discrepancy creates is presented in the standard error, which ranges from 10.894 to 12.553. This is the largest standard error of any variable. Overall, internet access demonstrates one of the largest and most consistent correlations with academic performance, establishing itself as one of the key determinants within the model.

Another significant variable is ‘homecomputer’, which maintains a statistically significant p-value across T*ables 1-3* and a positive coefficient ranging from 15.55 to 22.09. While this is still worth noting, it has a much lesser impact than ‘homeinternet’, suggesting that a computer itself is not of much significance without internet access. This data lines up with the conclusion of *Figure 2*, which depicted a higher median for those with a computer, but strong overlap with those without. The contrast between the significance of ‘homecomputer’ and ‘homeinternet’ can also be observed in the contrast between *Figure 2* and F*igure 3*, in which access to the internet is presented as a much more significant factor in academic performance. Overall, a home computer is correlated with improved learning outcomes, although its impact is modest compared to other variables.

Some variables, such as ‘quietplace’ and ‘wealth’, do not appear to be statistically significant, as demonstrated by their high p-values. The likely explanation for this is that what they contribute is already accounted for in other variables; for example, ‘room’ likely serves as a quiet place, making the ‘quietplace’ variable somewhat redundant. Regarding ‘wealth’, it seems counterintuitive that it is so negligible, but this is likely explained by the fact that variables such as ‘homeinternet’ and ‘homecomputer’ account for what having wealth would provide, namely home educational resources. As a result of this, the impact of wealth may already be captured indirectly through other variables, reducing the independent significance of ‘wealth’ as an explanatory variable- possibly because of multicollinearity between variables.

In conclusion, access to computers and other educational resources at home does impact pupils’ learning outcomes. Certain educational resources have more of an impact than others, with these models determining home internet access to be the most significant individual factor. Having books in the home was also established to be a significant factor, though this could be due to the wider socioeconomic implications of a family having books in the household as opposed to the books themselves as educational resources. Having computers in the house is another variable of significance. The models concluded that other variables also had a positive correlation with learning outcomes, though to a much lesser extent than the three mentioned. While not explicitly causal, this data implies a strong correlation between educational resources in the home and a pupil’s academic performance.

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